

Jinay Patel

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EDUCATION

University of Waterloo

Bachelor of Computer Engineering

Expected Graduation: Jun 2031

SKILLS

Languages: C/C++ (5 years), JavaScript/TypeScript (6 years), Python (5 years), Java (2 years)

Technologies: PyTorch, LangChain, Mastra, CMake, vcpkg, React, Svelte, Next JS, Tailwind CSS, Git, Docker, Unity, Linux

EXPERIENCE

Lead Software Engineer

October 2023 — June 2026

Seaquam Robotics

Delta, BC

- Built a data backend with Deno and TypeScript to aggregate data from 3 sources, cutting manual data entry by 50%
- Created a custom React and Tauri desktop CMS app, reducing data entry time from weeks to 60 minutes
- Built a Deno backend caching layer for slow third-party API endpoints, increasing server **response speed by 150%**

Embedded Controls Software Engineer

October 2021 — June 2023

Seaquam Robotics

Delta, BC

- Implemented C++ motion algorithms, securing a **100% win rate** during autonomous periods across 15+ matches
- Designed an interpolated lookup table using a monotonic cubic spline to tune PID constants in real time, increasing movement accuracy by 200% with **error tolerances under 1 inch**

Software Engineering Tutor

September 2025 — March 2026

Code Ninjas

Surrey, BC

- Helped 5 senior Black Belt students debug and implement features for their games being built with C# and Unity
- Designed ad campaigns with visual assets for Code Ninjas, improving the local branch's search visibility

Freelance Systems Administrator / IT Consultant

Convergence Media

Remote

- Created a custom Linux Mint ISO using Cubic with 3 bash scripts to automate the installation and setup of programs
- Deployed a self-hosted Nextcloud server (Microsoft 365 alternative) using Docker, which allowed file sharing and private video calls
- Configured a **Tailscale tunnel** to securely expose the Nextcloud server for remote video calls and file sharing

PROJECTS

Motion Control Framework | Embedded Controls Software Engineer

[GitHub](#)

- Used C++ to enable autonomous robot movement using PID controllers, feedforward controllers, and 2D motion profiles
- Designed a custom **2D motion profile generator** using cubic Bezier curves to calculate the robot's position, velocity, and acceleration at each point on the path while compensating for centripetal forces to keep the robot on the path
- Built **real-time visualization software** with Svelte and WebSockets to display live position, PID data, and logs

Verbosity: Custom Programming Language | Personal Project

[GitHub](#), [Blog Post](#)

- Built a custom lexer, parser, and code generator from scratch for a custom programming language using C++ and CMake
- Applied C++26 features such as abstract classes, virtual methods, smart pointers, and variants for the compiler pipeline

Termicord - Real-Time CLI Chat Application | Personal Project

[GitHub](#)

- Developed a Discord-like chat application which runs in the terminal using C++, raw TCP sockets, NCurses (for the TUI), and vcpkg
- Implemented a **custom HTTP server** with routing to handle cURL requests sent to a REST API along with a custom database

AWARDS

Vex Robotics World Champion, RECF & Vex Robotics

April 2026

Top overall team among 20,000+ worldwide, recognized for excellence across robot design, programming, and competitive performance

Excellence Award @ Provincial Championship, Vex Robotics

February 2026

Awarded to the top team out of the 100+ teams in British Columbia for excellent performance

VSHacks Hackathon 2nd Place, VSHacks

July 2025

Awarded for developing [Posturai](#), a real-time posture correction app

Robotics Programming Award x3, Waterloo CEMC

May 2026

Awarded for innovative motion control algorithms and technical documentation

Senior Computer Programming Award, Seaquam Secondary School

June 2026